

TrendCast

A multi-source, multimodal trend forecasting and scenario planning system

TrendCast is a multi-source trend intelligence system that integrates signals from NewsAPI, Reddit (public subreddit data), and Spotify. The system analyzes text, engagement, sentiment, and temporal patterns to identify trending topics, measure their novelty, forecast future momentum, and evaluate how trends evolve under different external scenarios.

Trend intelligence should not rely on a single platform or metric. TrendCast combines heterogeneous data sources, normalizes them for fair comparison, and identifies not only what is currently trending, but what is emerging and likely to gain momentum.

Team Chicago

Ishani Ashok Kumar
Parth Parag Shah
Deepti Ashok Agrawal
Mitali Kamal Bagadia
Nissie Mallyala

Key Challenges in Trend Detection

Cultural trends behave differently across platforms, making single-source tracking incomplete and often misleading.

Why not use one source?

News platforms capture institutional attention, Reddit reflects community discussions, and Spotify captures entertainment and audio-cultural behavior. Each platform represents a different signal, so relying on a single source provides only a partial and biased view of trends.

Why is raw engagement not enough?

Engagement metrics differ across platforms (e.g., Reddit upvotes, Spotify popularity, news mentions). Direct comparison is misleading as the noisiest source wins. TrendCast normalizes within each source before combining signals.

Why add novelty?

A topic can be popular but not new. Traditional systems fail to distinguish between established demand and emerging trends. TrendCast introduces novelty scoring to identify topics that are both gaining attention and deviating from baseline patterns.

Project goal

How can signals from multiple online platforms be combined into a unified and interpretable system for detecting, ranking, forecasting, and understanding emerging trends?

TrendCast addresses this by integrating data from multiple sources, standardizing engagement across platforms, and applying sentiment and novelty analysis. The system produces clear, explainable trend scores and presents insights through an interactive dashboard. This enables users to move beyond simple trend tracking toward deeper, data-driven insights.

TrendCast Solution Overview

Multi-Source Integration paragraph text

- TrendCast combines data from News, Reddit, and Spotify to capture different perspectives of the same trend.
- Each platform reflects a different type of signal news shows institutional attention, Reddit captures community discussion, and Spotify reflects entertainment behavior.
- By bringing these together, the system avoids the bias of single-source tracking and creates a more complete view of real-world trends.

Interpretable Trend Intelligence

- TrendCast identifies what is trending, distinguishes emerging topics from established ones, and tracks how trends evolve over time.
- It produces clear, explainable outputs including trend scores, novelty scores, and forecasts so users understand not just what is happening, but why.
- This enables more informed, data-driven decisions instead of relying on raw engagement metrics.

Multi-Source Data Pipeline

NewsAPI

Why use it:

Shows media coverage and institutional attention.

How used:

Search by canonical topic query and collect titles, descriptions, dates, and image URLs.

Spotify

Why use it:

Adds entertainment and audio-cultural signals.

How used:

Search tracks by topic, collect popularity, album art, and audio features when available.

Reddit

Why use it:

Captures early community discussion and niche interest before mainstream coverage.

How used:

Search by topic terms inside selected subreddits or sitewide.

Optional X/Twitter

Why use it:

Adds fast-moving public reaction.

How used:

Only if a bearer token or CSV export is provided, so the system remains usable without it.

Why subreddits?

Subreddits act like community filters. Searching inside relevant communities reduces noise and makes the collected posts more meaningful. For example, a topic can be mapped to focused spaces such as technology, music, mindfulness, or fashion rather than pulling everything from general Reddit.

TrendCast also supports '*' for broader sitewide Reddit search when a niche community filter is too restrictive.

How Reddit is handled robustly

Reddit data collection is designed to be resilient to API limits and noisy inputs.

The system retries failed requests, controls request rates, and removes duplicate posts to maintain data quality.

Instead of failing when limits are hit, it supports partial data collection so the pipeline continues to run and still produces usable results.

How the system works end to end

1. Canonical topics

The system begins with saved canonical topics or user-entered search topics

2. Data Collection

Data is collected from News, Reddit, and Spotify, where each source captures a different type of signal.

3. Processing

The data is cleaned, aligned, and transformed through sentiment extraction and normalization to make sources comparable.

4. Scoring

Each topic is evaluated using trend strength, sentiment, and novelty to capture both popularity and emerging signals.

5. Enrichment

Additional signals such as topic patterns, temporal trends, and forecasting are used to deepen the analysis.

6. Insights Layer

The final output presents ranked, explainable insights through an interactive dashboard.

How TrendCast Ranks Topics

Popularity Signals

- Measures current topic strength
- Uses mention volume
- Uses normalized engagement
- Uses sentiment
- Captures attention, response, and tone

Emerging Signal

- Uses a KL-divergence-based novelty score
- Compares topic prominence to the rest of the topic set
- Highlights unusually prominent topics
- Separates established popularity from emerging signals

Combined Score

- Combines popularity and novelty into one ranking
- Balances current relevance with emerging momentum
- Highlights strong established topics
- Also surfaces rising opportunities

Why This Works

- Normalizes engagement within each source
- Prevents one platform from dominating the ranking
- Makes cross-source comparison fairer
- Creates a more balanced result

Final Output

- Produces a ranked topic list
- Includes interpretable scores
- Includes novelty signals
- Supports dashboard-ready visuals
- Helps users compare topics and understand ranking drivers

Why BERTopic Was Used for Topic Discovery

Why topic modeling at all?

The project collects a large amount of text from news, Reddit, and other sources. Looking at individual posts one by one is noisy and hard to interpret. Topic modeling helps group similar documents together so we can identify the main themes people are discussing.

Why BERTopic specifically?

BERTopic was chosen because it creates topic clusters with readable keywords, which makes the results easier to interpret and present. It is stronger than simple keyword counting because it groups related language into themes instead of only counting repeated words.

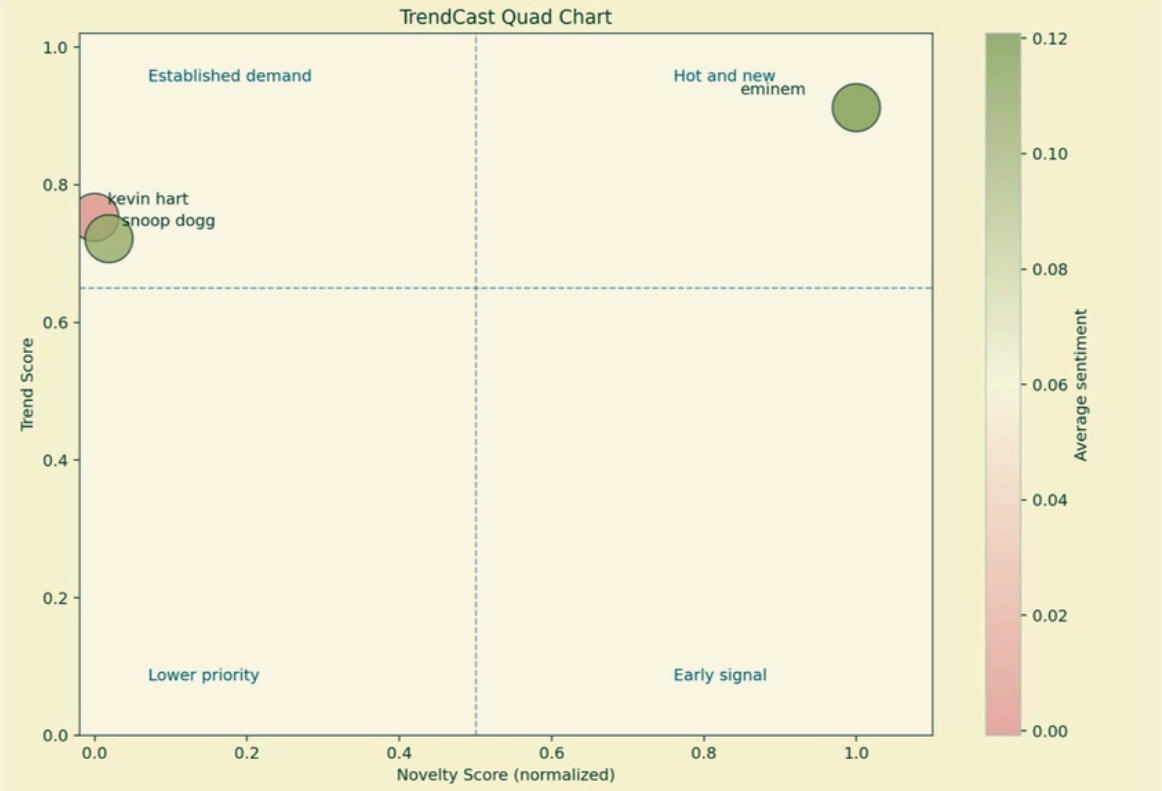
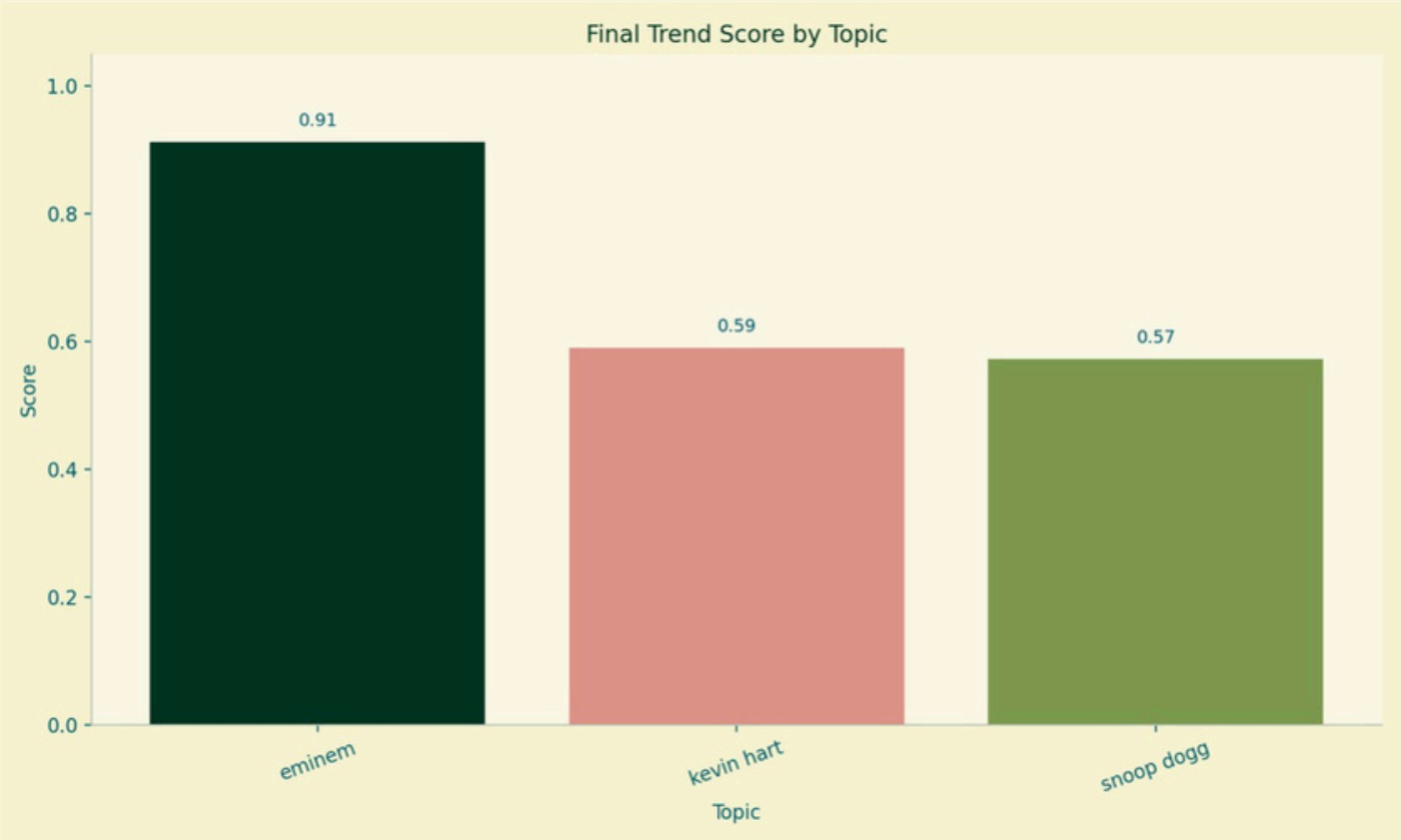
Why is there a fallback?

Capstone projects must stay runnable on different machines. If BERTopic is unavailable, the project falls back to a local BERTopic-style pipeline using TF-IDF, dimensionality reduction, and clustering so the workflow still works.

How to explain BERTopic in one sentence

BERTopic helps us discover the main themes inside the collected text and summarize them with readable keywords, so the system can say not only which topic is winning, but also what language and ideas are driving that topic underneath the surface.

What the stable demo outputs show



Key result

Eminem achieved a score of 0.91, substantially ahead of Kevin Hart at 0.59 and Snoop Dogg at 0.57. The 32-point margin reflects stronger combined evidence across mention volume, engagement, sentiment, and novelty dimensions.

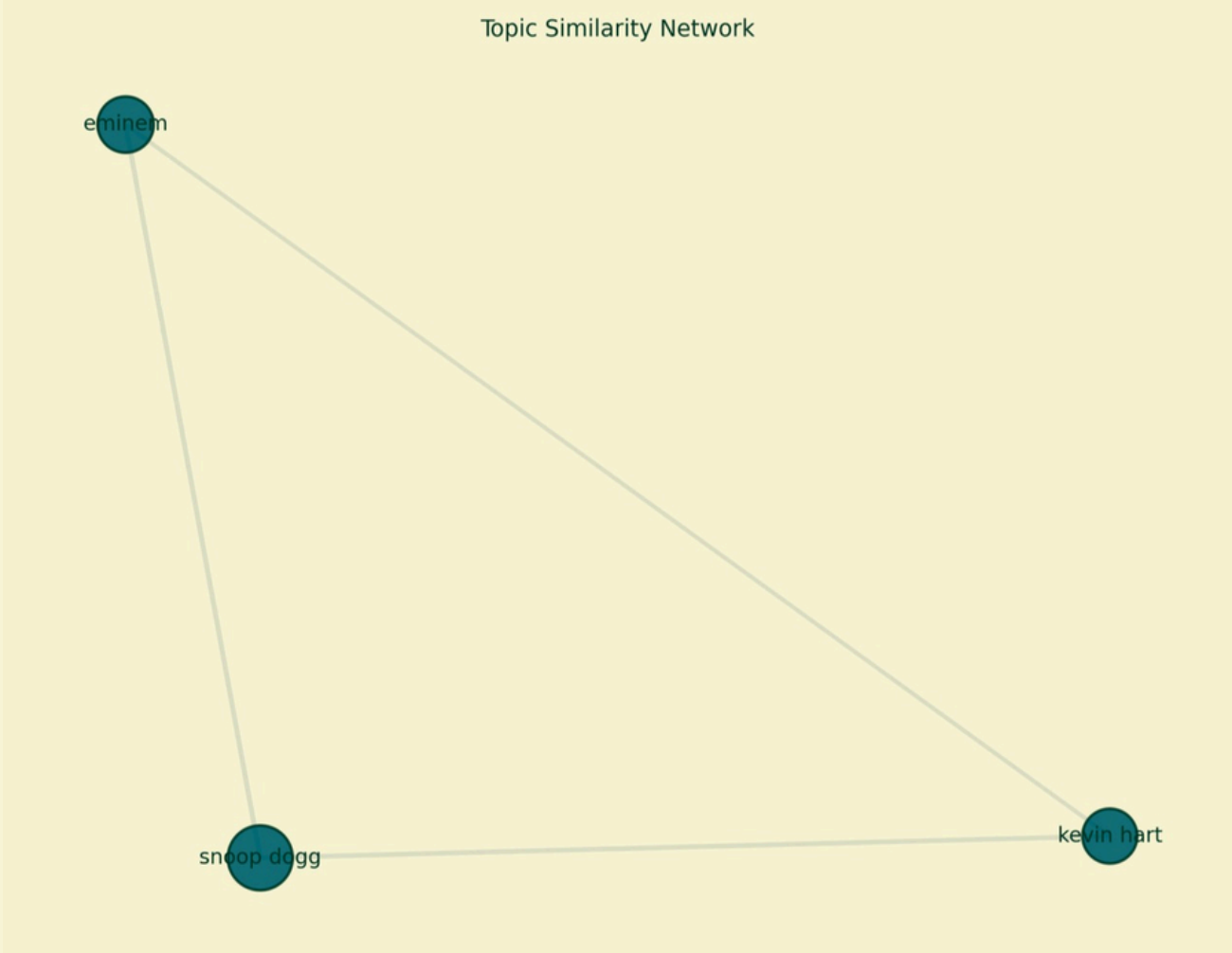
Trend vs novelty insight

Eminem appeared in the high-trend/high-novelty quadrant, while Kevin Hart and Snoop Dogg showed stable attention but lower novelty.

Best one-line interpretation

TrendCast successfully distinguishes emerging trends from established popularity using multi-source signals.

Why TrendCast Is a Decision-Support System



Forecasting

TrendCast estimates near-term momentum using an ensemble of Bayesian, Holt, and momentum methods.

Scenario Planning

TrendCast simulates how topic rankings may change under external scenarios such as creator economy growth, mental-health-priority shifts, recession shock, or AI regulation.

Explainability + Dashboard

TrendCast explains which signals are driving rankings and presents them through an interactive dashboard, making the output easier to interpret and act on.

Six Areas for Future Development

Official Reddit API Integration

Transition from public data retrieval to authenticated API access for structured metadata, improved reliability, and higher request throughput. (Rate limit challenge)

Twitter Data Inclusion

Incorporate real-time public reaction signals from X to address the coverage gap in the data pipeline. (Expensive)

Extended Forecasting Horizon

Collect longer historical time series to support LSTM and Transformer-based forecasters, enabling more robust and accurate predictions.

Ground Truth Outcome Validation

Integrate external signals such as search volume growth, streaming activity, and event participation to enable supervised learning in the fusion model.

Data Driven Scenario Calibration

Learn scenario impact weights from documented historical events such as regulatory announcements, economic contractions, and cultural shifts.

Domain Specific Expansion

Extend TrendCast beyond entertainment into finance, healthcare, technology, and marketing analytics with domain-appropriate data sources.

From heterogeneous data to actionable intelligence

TrendCast moves beyond descriptive monitoring. It shows how heterogeneous online signals can be fused into an interpretable system for understanding what is trending, what is emerging, and what may happen next.